

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

20 September 2018

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper

Section B

Answer any **one** question on the separate Answer Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

This document consists of **14** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Malaria is an infectious disease caused by *Plasmodium*. *Plasmodium* requires two hosts to complete its complex life cycle. One of the hosts is the *Anopheles* mosquito, which acts as a vector of malaria.

Transmission of malaria occurs when females of some species of *Anopheles* take blood meals from humans infected with *Plasmodium*, and then feed on uninfected individuals.

Both male and female *Anopheles* mosquitos have piercing and sucking mouthparts. The female mosquito is shown in **Fig. 1.1**.

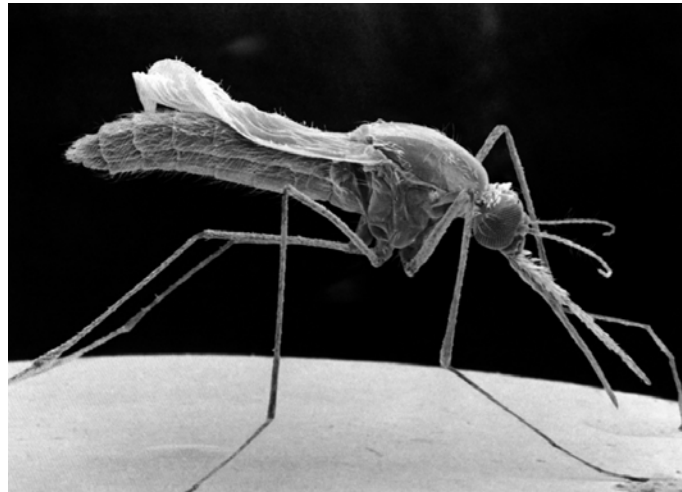


Fig.1.1.

- (a) *Plasmodium* are responsible for millions of infections each year in tropical and subtropical areas of the world. According to the World Health Organization, malaria incidence has increased significantly over the past 50 years, turning this infection into the most important mosquito-borne disease in the world and a global health challenge.

Fig. 1.2 shows the general global distribution of malaria in the year 2005.

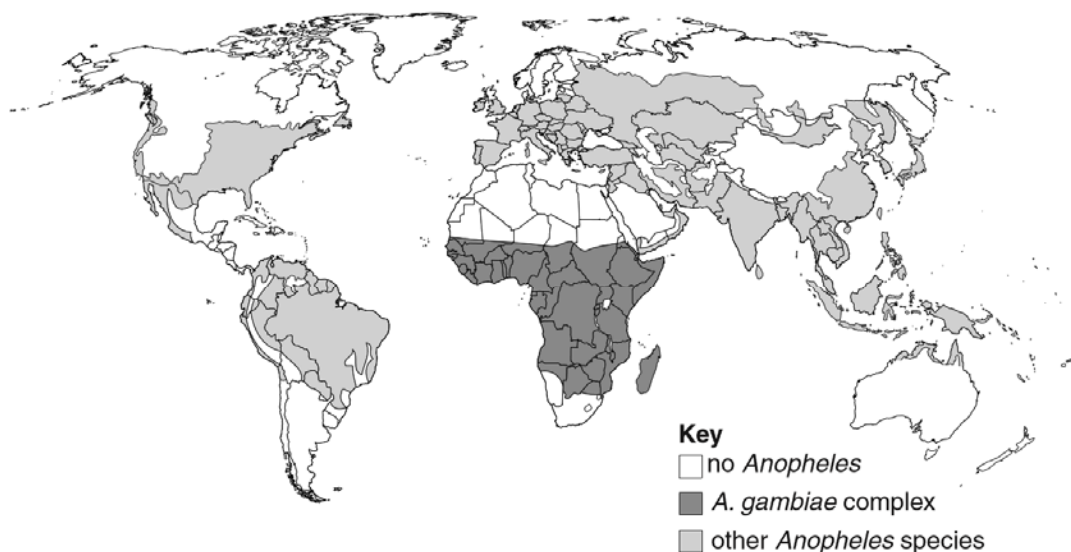


Fig.1.2.

- (i) Climate is one of the important factors that affects the distribution of malaria.

Predict and justify the expected distribution of malaria by the end of the 21st century.

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- (ii) Symptoms of malaria such as high fever usually develop within 7 to 18 days after being bitten by an infected mosquito and is often associated with muscle pain.

Suggest how the *Plasmodium* may cause these symptoms.

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During an infection, *Plasmodium* will elicit humoral immune response which involves B cell activation as shown in **Fig.1.3**.

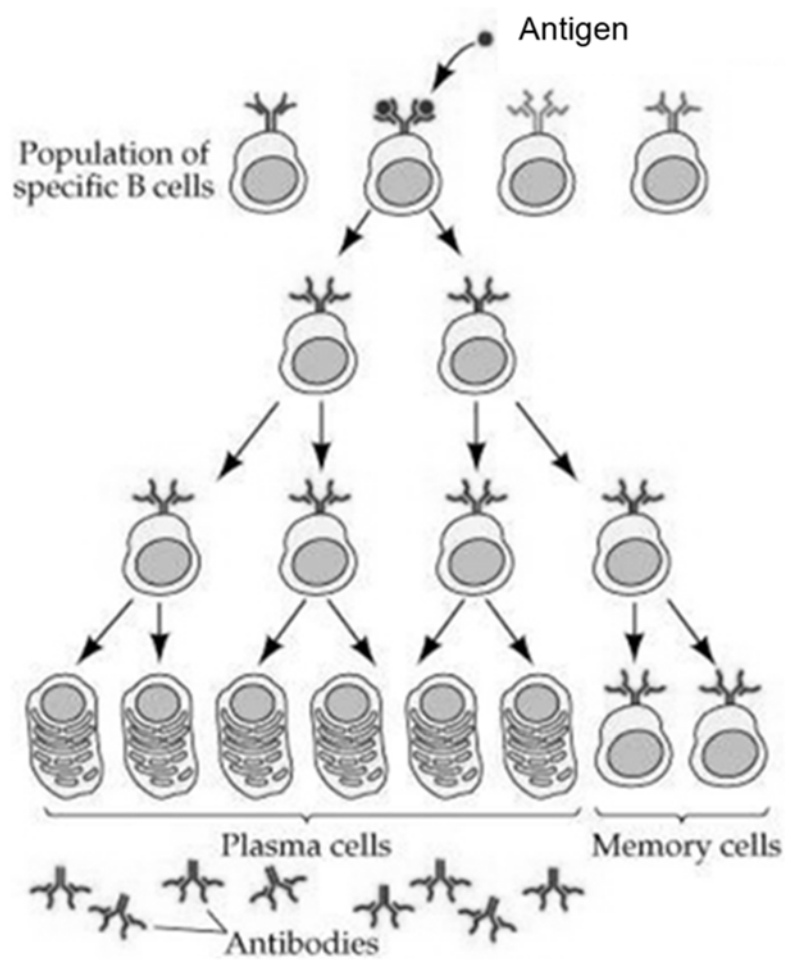


Fig.1.3.

(iii) With reference to **Fig. 1.3** and your own knowledge, explain the significance of mitosis in B cell activation.

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- (iv) Memory B cells have long life spans because they do not actively undergo cell division. However, upon activation, a memory B cell can undergo many round of proliferation.

Suggest why activated memory B cells can undergo multiple rounds of cell division without dying.

[1]

Fig. 1.3 is part of a complex food web in an area of Kenya where the larvae and adults of *A. gambiae* occur.

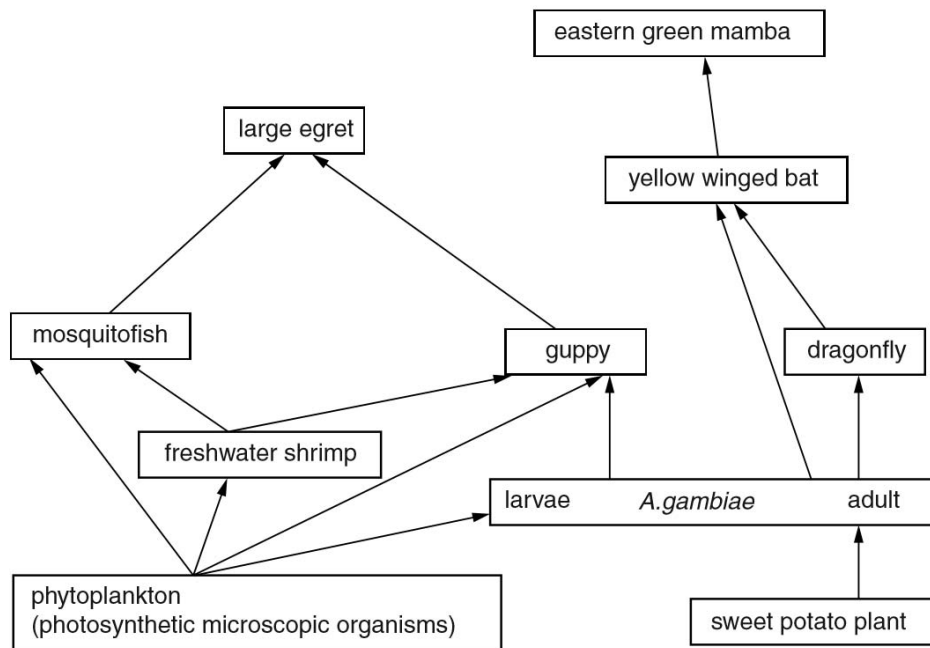


Fig.1.3

- (v) Suggest how the information in **Fig. 1.3** can be used in the control of malaria in other areas of Kenya.

[2]

- (c)** Each molecule of catalase consists of four identical polypeptides. The two forms of catalase in *A. gambiae* differ by only one amino acid at position 2 in the amino acid sequence. Catalase P has serine and catalase Q has tryptophan.

Suggest how the difference in one amino acid is responsible for the lower activity of catalase Q compared with catalase P.

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[2]

- (d)** Female mosquitoes feed on blood in order to produce their eggs. After feeding, the metabolic rate increases for egg production.

The researchers allowed female mosquitoes to feed on blood. They found that female mosquitoes with only catalase P produced more eggs than those with only catalase Q.

Suggest why there is a difference in egg production between the two types of *A. gambiae*.

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- (e)** Copper ions can act as a non-competitive inhibitor of catalase.

Explain how copper ions can be used in the fight against malaria.

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- (f) Malaria is an important disease worldwide. **Table 1.1** shows recent information about malaria cases reported during one year in six different countries.

Table. 1.1

country	region	number of cases	number of cases per 100 000 population
Germany	Europe	4000	5
India	Asia	2 300 000	185
Japan	Asia	27 000	21
South Africa	Africa	490 000	981
Swaziland	Africa	15 000	1287
United Kingdom	Europe	7900	13

With reference to Table 1.1, explain the advantage of calculating the number of cases of malaria per 100 000 population rather than stating the number of cases alone.

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- (g) Suggest why malaria is more likely to be fatal in people who have HIV/AIDS than in those who do not have HIV/AIDS.

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[Total: 29]

- 2 In anaerobic respiration in yeast, the pyruvate molecules are broken down to produce ethanol and carbon dioxide. The release of carbon dioxide can be used to investigate the rate of anaerobic respiration.

Fig. 2.1 shows an experiment which was set up to find the rate of anaerobic respiration.

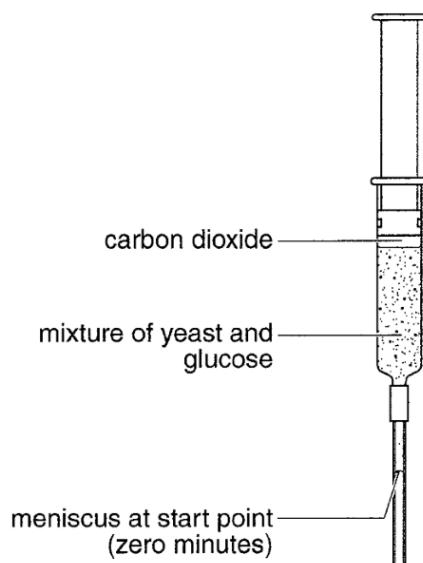


Fig. 2.1

The meniscus moves down the tube as carbon dioxide is released.

Table 2.1 shows the distance moved by the meniscus from the start point. This was recorded every 10 minutes.

Table 2.1

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	1	2	5	9	14	21	45	73	98

The rate of anaerobic respiration can be calculated by using the rate of movement of the meniscus.

Calculate the rate of anaerobic respiration between 70 and 80 minutes.

You will lose marks if you do not show your working.

Rate of anaerobic respiration mm min⁻¹ [2]

(a) The experiment was repeated three more times. Each time, the glucose (a monosaccharide) was replaced with a different disaccharide sugar:

- Maltose – a disaccharide of glucose and glucose.
- Sucrose – a disaccharide of glucose and fructose.
- Lactose – a disaccharide of glucose and galactose.

Table 2.2 (a), (b) and (c) show the results of these experiments.

Table 2.2 (a): Using maltose

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	0	0	0	0	2	3	6	9	12

Table 2.2 (b): Using sucrose

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	0	0	1	3	11	22	37	48	61

Table 2.2 (c): Using lactose

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	0	0	0	0	0	0	0	0	0

With reference to the information provided in Table 2.2 (a), (b) and (c) and your biological knowledge:

- (i) describe the difference in the results for maltose and sucrose, and suggest one explanation for this difference,

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(ii) Suggest **two** explanations for the results for lactose.

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(b) During anaerobic respiration in mammals, pyruvate is broken down by a different pathway from that in yeast.

Outline what happens to pyruvate in mammals during anaerobic respiration.

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[Total: 10]

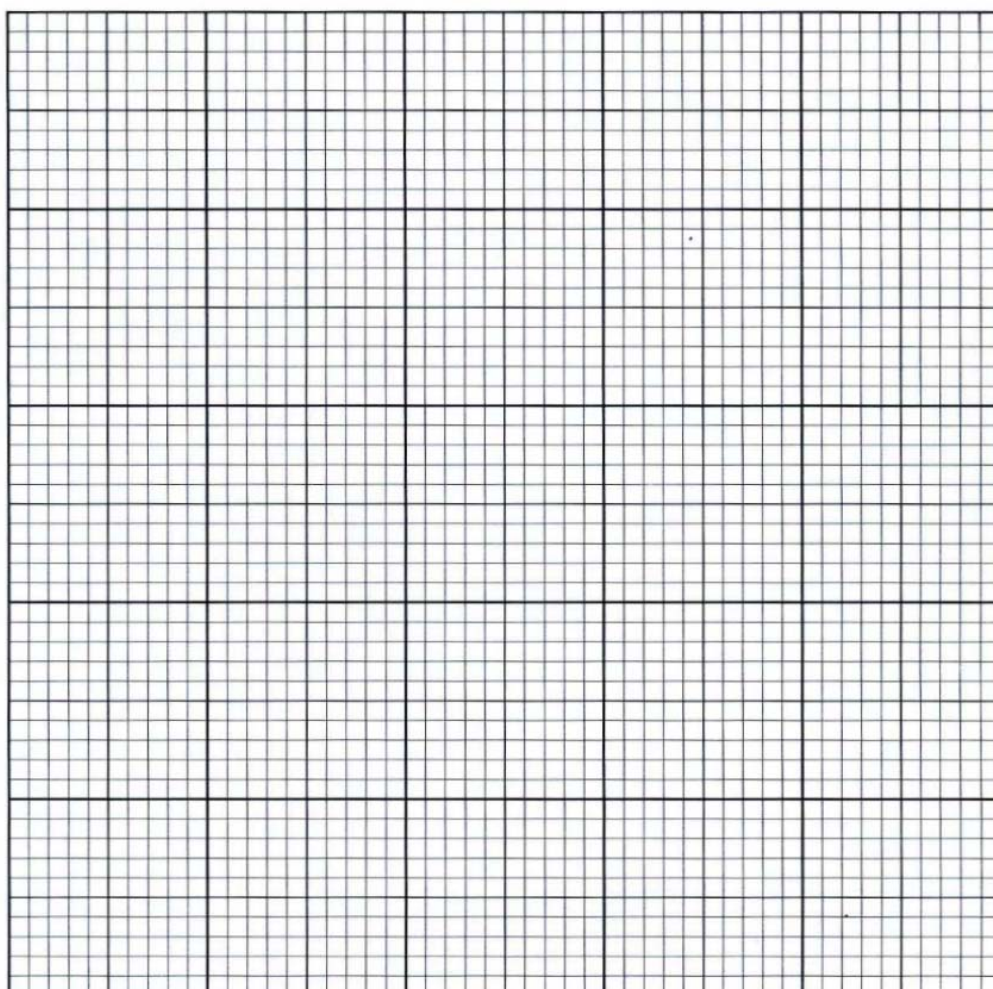
- 3 **Table 3.1** shows the results from an investigation to determine how two different drugs, **A** and **B**, enter animal cells.

Cells which did not contain either drug were placed into separate containers. Different concentrations of each drug were added and after 5 minutes the drug concentration in the cell was measured.

Table 3.1

concentration of drug A in container /arbitrary units	concentration of drug A inside the cells after 5 minutes /arbitrary units	concentration of drug B in container /arbitrary units	concentration of drug B inside the cells after 5 minutes /arbitrary units
0	0	0	0
5	5	5	4
10	10	10	7
15	15	15	11
20	19	20	13
25	24	25	13

- (a) Plot a graph of these data on the same pair of axes.



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- (b)** From your knowledge of the cell surface membrane, explain if the information in **Table 3.1** suggests that drug A is lipid soluble.

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- (c)** Suggest, with reasons, how drug **B** is transported into the cells.

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[Total: 11]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4(a)** The cell theory is based on a number of principles. [9]
Discuss the extent, in humans, to which these principles can be accounted for by mitosis and the different types of stem cells.
You should consider both individual humans and populations of humans from one generation to the next.
- (b)** Outline the roles of hydrogen bonds in biomolecules. [6]
- (c)** Discuss the suggestion that all living organisms on Earth depend on phosphate. [10]

[Total: 25]

- 5(a)** In 2014, Dalrymple, Buswell and Moles published an articles on their studies of plants and their proteins, after being introduced to New Zealand and Australia. [6]
They concluded that asexually reproducing species change as rapidly as sexually reproducing species when introduced to a new environment. This conclusion is unexpected.
Explain why this conclusion is unexpected.
- (b)** During the study of proteins, explain how it is possible for both plant amylase and bacterial amylase to break down the same substrate, but for the bacterial amylase to have a higher optimum temperature. [6]
- (c)** A comparison of DNA sequences of many genes from a variety of species has shown that cytochrome c is an example of molecular homology. [13]
Gel electrophoresis can be used to compare DNA that codes for cytochrome c.
Describe and explain how gel electrophoresis can be used to compare DNA that codes for cytochrome c to show the evolutionary relationships between a variety of species.

[Total: 25]